Question 1

```python
# Q1: Sum all items in a list
numbers = [1, 2, 3, 4]
total = 0
for n in numbers:
    total += n
print("Sum:", total)
```

Question 2

```python
# Q2: Largest number from a list
numbers = [1, 9, 3, 7]
largest = numbers[0]
for n in numbers:
    if n > largest:
        largest = n
print("Largest:", largest)
```

Question 3

```python
# Q3: Smallest number from a list
numbers = [5, 2, 8, 1]
smallest = numbers[0]
for n in numbers:
    if n < smallest:
        smallest = n
print("Smallest:", smallest)
```

Question 4

```python
# Q4: Display first and last colors
color_list = ["Red","Green","White","Black"]
print("First Color:", color_list[0])
print("Last Color:", color_list[-1])
```

Question 5

```python
# Q5: Add 'ing' or 'ly' to string
s = "play"
if len(s) >= 3:
    if s.endswith("ing"):
        s = s + "ly"
    else:
        s = s + "ing"
print(s)

s = "playing"
if len(s) >= 3:
    if s.endswith("ing"):
        s = s + "ly"
    else:
        s = s + "ing"
print(s)
```

Question 6

```python
# Q6: Student division based on marks
marks = [60, 70, 80, 90, 100]
percentage = sum(marks) / len(marks)

if percentage >= 60:
    print("First Division")
elif percentage >= 50:
    print("Second Division")
elif percentage >= 40:
    print("Third Division")
else:
    print("Fail")
```

Question 7

```python
# Q7: Largest among three numbers
a, b, c = 10, 25, 15
if a >= b and a >= c:
    largest = a
elif b >= a and b >= c:
    largest = b
else:
    largest = c
print("Largest:", largest)
```

Question 8

```python
# Q8: Check leap year
year = 2024
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

Question 9

```python
# Q9: Palindrome check
s = "madam"
if s == s[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Question 10

```python
# Q10: Extend nested list
list1 = ["a", "b", ["c", ["d", "e", ["f", "g"], "k"], "l"], "m", "n"]
sub_list = ["h", "i", "j"]
list1[2][1][2].extend(sub_list)
print(list1)
```

Question 11

```python
# Q11: Replace first 20 with 200
list1 = [5, 10, 15, 20, 25, 50, 20]
for i in range(len(list1)):
    if list1[i] == 20:
        list1[i] = 200
        break
print(list1)
```

Question 12

```python
# Q12: Rotate list to the right by k steps
lst = [1, 2, 3, 4, 5]
k = 2
k = k % len(lst)
rotated = lst[-k:] + lst[:-k]
print(rotated)
```

Question 13

```python
import os
import re
from pygments import highlight
from pygments.lexers import PythonLexer   # □ Python lexer
from pygments.formatters import ImageFormatter
from PIL import Image
from reportlab.pdfgen import canvas
from reportlab.lib.pagesizes import A4
from docx import Document

# -------------------------
# Natural sort key (so q2.py comes before q10.py)
# -------------------------
def natural_sort_key(text):
    return [int(s) if s.isdigit() else s.lower() for s in re.split(r'(\d+)', text)]

# -------------------------
# 1. Collect all .py files in natural order
# -------------------------
code_files = [f for f in os.listdir() if f.endswith(".py")]
code_files.sort(key=natural_sort_key)

if not code_files:
    print("□□ No .py files found in this directory.")
    exit()

# -------------------------
# 2. Create screenshots of each code file (dark mode)
# -------------------------
images = []
for i, filename in enumerate(code_files, start=1):
    with open(filename, "r", encoding="utf-8") as f:
        code = f.read()

    formatter = ImageFormatter(
        font_name="Consolas",   # safe on Windows, fallback to DejaVu Sans Mono on Linux
        font_size=18,
        line_numbers=True,
        style="monokai"  # dark mode theme
    )
    img_data = highlight(code, PythonLexer(), formatter)  # □ use PythonLexer

    img_name = f"question{i}.png"
    with open(img_name, "wb") as img_file:
        img_file.write(img_data)

    images.append((i, img_name))
    print(f"□ Created screenshot: {img_name} from {filename}")

# -------------------------
# 3. Make PDF from images
# -------------------------
pdf_file = "Questions.pdf"
c = canvas.Canvas(pdf_file, pagesize=A4)
width, height = A4

for i, (q_num, img_name) in enumerate(images, start=1):
    # Heading
    c.setFont("Helvetica-Bold", 18)
    c.drawString(200, height - 50, f"Question {q_num}")

    try:
        img = Image.open(img_name)
        aspect = img.width / img.height

        new_width = width - 80
        new_height = new_width / aspect
        if new_height > height - 150:
            new_height = height - 150
            new_width = new_height * aspect

        c.drawImage(img_name, 40, height - new_height - 100, new_width, new_height)
    except Exception as e:
        print(f"□□ Could not add {img_name}: {e}")

    c.showPage()

c.save()
print("□ PDF created:", pdf_file)

# -------------------------
# 4. Make Word file with code text
# -------------------------
doc = Document()
for i, filename in enumerate(code_files, start=1):
    doc.add_heading(f"Question {i}", level=1)
    with open(filename, "r", encoding="utf-8") as f:
        code = f.read()
    doc.add_paragraph(code, style="Normal")

word_file = "Questions.docx"
doc.save(word_file)
print("□ Word file created:", word_file)
```